

Nộp bài tập homework ss11

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- Why does setup timing show violations after routing even though it was clean after postCTS?
 - Ở giai đoạn postCTS đã clean setup timing nhưng sau khi routing lại bị violate setup là vì trước routing/postCTS tín hiệu kết nối với nhau là giả định (virtual connect), delay của signal và RC delay đều được tool ước lượng tính toán, nhưng sau khi route, các signal đều được kết nối với nhau bằng metal thật, tool sẽ dựa trên metal thật đó mà tính delay.
 - Việc setup timing bị violate ở routing có thể là do data path tăng delay do detour, hay routing congestion, nếu hai dây song song quá dài thì có thể dẫn đến coupling, crosstalk giữa các net khiến nó tăng delay -> violate setup timing.

2. Fix DRC theo yêu cầu HW

- Run: `verify_drc -limit 0`

```
Verification Complete : 11 Viols.

Violation Summary By Layer and Type:

      Short  MetSpc  Totals
Metal2      2       0       2
Metal3      1       0       1
Metal4      1       1       2
Metal5      2       4       6
Totals      6       5      11

*** End Verify DRC (CPU: 0:00:11.3  ELAPSED TIME: 12.00  MEM: 57.0M) ***
```

- Có 11 violate, có 6 violate short và 5 violate Length spacing

Violation Type:		Violation:		
NanoRoute (5/5)		LAYER	OBJECT1	OBJECT2
Process Antenna (5/5)		Metal4(4)	i_croc_soc/CTS_4	
Antenna_Cumulative_Area_Ratio (5/5)		Metal4(4)	i_croc_soc/CTS_7	
Metal4(4) (5/5)		Metal4(4)	i_croc_soc/i_croc/i_core_wra...	
Verify (11/11)		Metal4(4)	i_croc_soc/i_croc/i_core_wra...	
Geometry (11/11)		Metal4(4)	soc_gpio_i_25_	
Metal_Short (6/6)				
Metal2(2) (2/2)				
Metal3(3) (1/1)				
Metal4(4) (1/1)				
Metal5(5) (2/2)				
Parallel_Run_Length_Spacing (5/5)				
Metal4(4) (1/1)				
Metal5(5) (4/4)				

- Sau khi fix thủ công các dây bị short và length spacing thì đã clean violate
- Chạy lại lệnh: verify_drc -limit 0

```

VERIFY DRC ..... Sub-Area : 79 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1275.680 1457.920 1457.920 1640.000} 80 of 81
VERIFY DRC ..... Sub-Area : 80 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1457.920 1457.920 1640.000 1640.000} 81 of 81
VERIFY DRC ..... Sub-Area : 81 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:11.2  ELAPSED TIME: 11.00  MEM: 0.0M) ***

1
innovus 9>

```

- Chạy lệnh: verify_connectivity -type regular

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innovus 10> verify_connectivity -type regular
VERIFY_CONNECTIVITY use new engine.

***** Start: VERIFY CONNECTIVITY *****
Start Time: Thu Aug 13 14:27:49 2026

Design Name: croc_chip
Database Units: 1000
Design Boundary: (0.0000, 0.0000) (1640.0000, 1640.0000)
Error Limit = 1000; Warning Limit = 50
Check specified nets
Net i_croc_soc/i_croc/FE_OFN4067_n: dangling Wire.
**** 14:27:49 **** Processed 5000 nets.
Net i_croc_soc/i_croc/i_main_xbar_gen_demux_1__i_demux_i_counter_q_o: dangling Wire.
Net i_croc_soc/i_croc/i_core_wrap/FE_OCPN8022_2466: dangling Wire.
Net i_croc_soc/i_croc/i_core_wrap/FE_OCPN5434_FE_OFN1391_0852: dangling Wire.
Net i_croc_soc/i_croc/i_core_wrap/FE_RN_2330_0: dangling Wire.
**** 14:27:49 **** Processed 10000 nets.
Net i_croc_soc/i_croc/i_core_wrap/_2277_: dangling Wire.
**** 14:27:50 **** Processed 15000 nets.
Net i_croc_soc/i_croc/i_core_wrap/i_ibex_cs_registers_i/_1110_: dangling Wire.
**** 14:27:50 **** Processed 20000 nets.
**** 14:27:50 **** Processed 25000 nets.
Net i_croc_soc/i_croc/i_dm_top_i_dm_top/_0467_: has regular routing with opens, dangling Wire.
**** 14:27:50 **** Processed 30000 nets.
Net i_croc_soc/i_croc/i_dm_top_i_dm_top/_1986_: has regular routing with opens, dangling Wire.
**** 14:27:50 **** Processed 35000 nets.
**** 14:27:50 **** Processed 40000 nets.
**** 14:27:50 **** Processed 45000 nets.

Begin Summary
  5 Problem(s) (IMPVFC-92): Pieces of the net are not connected together.
  15 Problem(s) (IMPVFC-94): The net has dangling wire(s).
  20 total info(s) created.
End Summary

End Time: Thu Aug 13 14:27:51 2026
Time Elapsed: 0:00:02.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 20 Viols.  0 Wrngs.
(CPU Time: 0:00:01.4  MEM: 0.000M)

1
innovus 11> win
innovus 12>

```